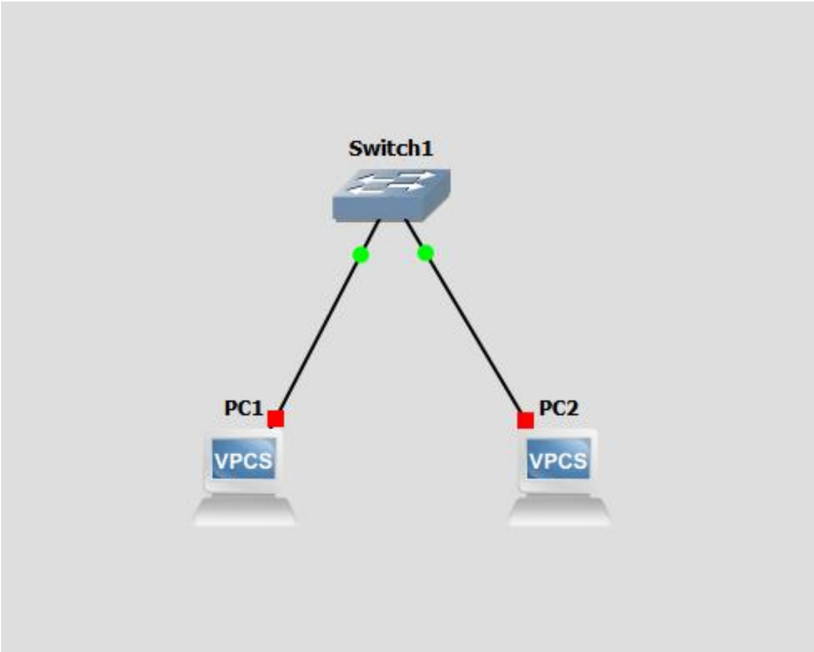
Лабораторная работа №1

Тема: Освоение инструментария для выполнения работ, построение простой сети

2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети.



Для PC1: ip 192.168.1.1 255.255.255.0

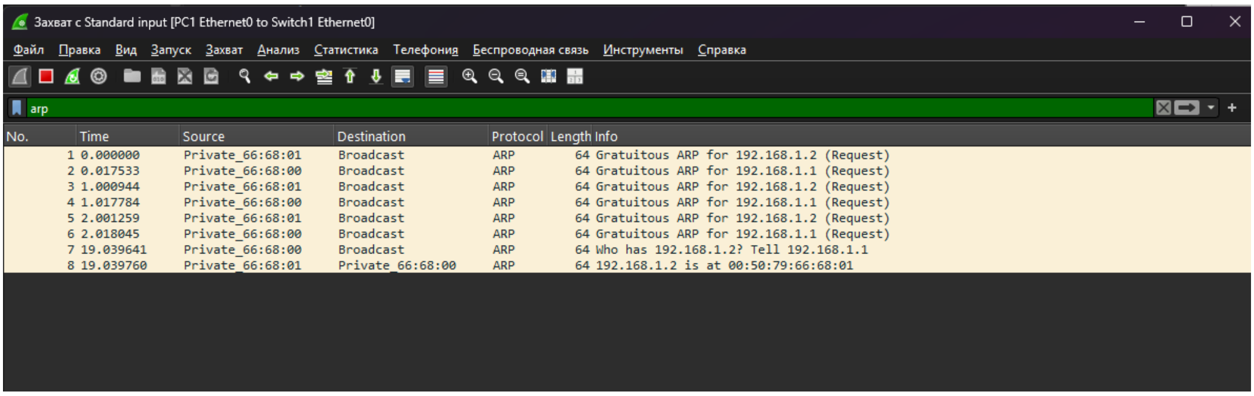
Для PC2: ip ip 192.168.1.2 255.255.255.0

3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера.

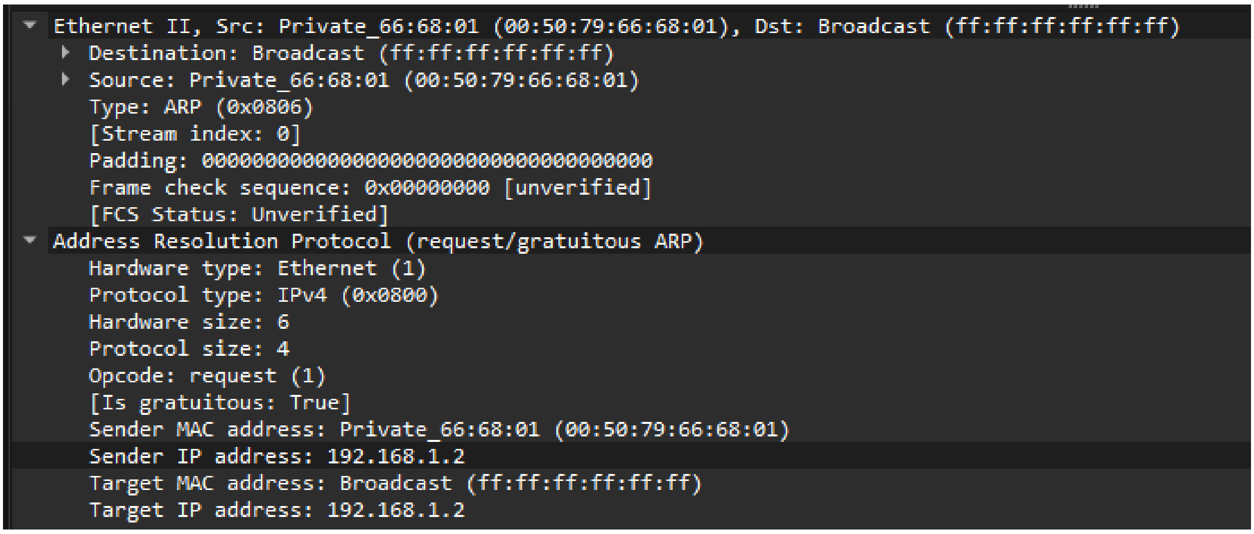
```
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=0.220 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=0.251 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.226 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=0.242 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=0.235 ms

PC1>
```

4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark.

Трафик на обоих линках одинаковый:



первые 6 ARP-пакетов (Gratuitous ARP), нужны для предотвращения конфликтов ip адресов в сети: если придет ответ, значит адрес занят.



широковещательный запрос (Dest: Broadcast ff:ff:ff:ff:ff:ff).

Sender ip и target ip совпадают, потому что устройству необходимо получить ответ от другого узла, который использует такой же ip.

Who has 192.168.1.2? Tell 192.168.1.1

```
Frame 7: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
  Destination: Broadcast (ff:ff:ff:ff:ff:ff)
    .... ..1. .... .... .... .... = LG bit: Locally administered address (this is NOT the factory defaul
    .... ...1 .... .... .... .... = IG bit: Group address (multicast/broadcast)
  Source: Private_66:68:00 (00:50:79:66:68:00)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
  Padding: 000000000000000000000000000000000000
  Frame check sequence: 0x00000000 [unverified]
  [FCS Status: Unverified]
Address Resolution Protocol (request)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: request (1)
  Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Sender IP address: 192.168.1.1
  Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
  Target IP address: 192.168.1.2
```

Также широковещательный запрос (Dest: Broadcast ff:ff:ff:ff:ff:ff)

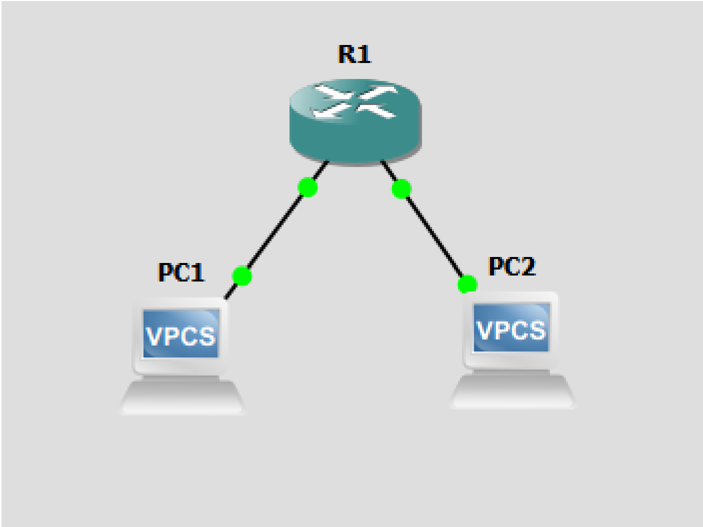
Target IP address: 192.168.1.2 – ip устройства MAC адресс которого нужно узнать.

ARP ответ:

```
Frame 8: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
  Destination: Private_66:68:00 (00:50:79:66:68:00)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: Private_66:68:01 (00:50:79:66:68:01)
    .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
    .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 2]
  Padding: 000000000000000000000000000000000000
  Frame check sequence: 0x00000000 [unverified]
  [FCS Status: Unverified]
Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.1.2
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.1.1
```

Sender MAC address: Private_66:68:01 (00:50:79:66:68:01 – искомый MAC-адрес)

5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей.



PC1: ip 192.168.1.10 255.255.255.0 192.168.1.1

PC2: ip 10.0.0.10 255.255.255.0 10.0.0.1

R1:

```
R1#
R1#Q
R1#enable
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface fastEthernet 0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Mar  1 00:03:34.371: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:03:35.371: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config)#interface fastEthernet 1/0
R1(config-if)#ip address 10.0.0.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar  1 00:03:59.635: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 00:04:00.635: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit
R1(config)#end
R1#
*Mar  1 00:04:19.515: %SYS-5-CONFIG_I: Configured from console by console
R1#
```

enable

configure terminal

interface fastEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface fastEthernet 1/0

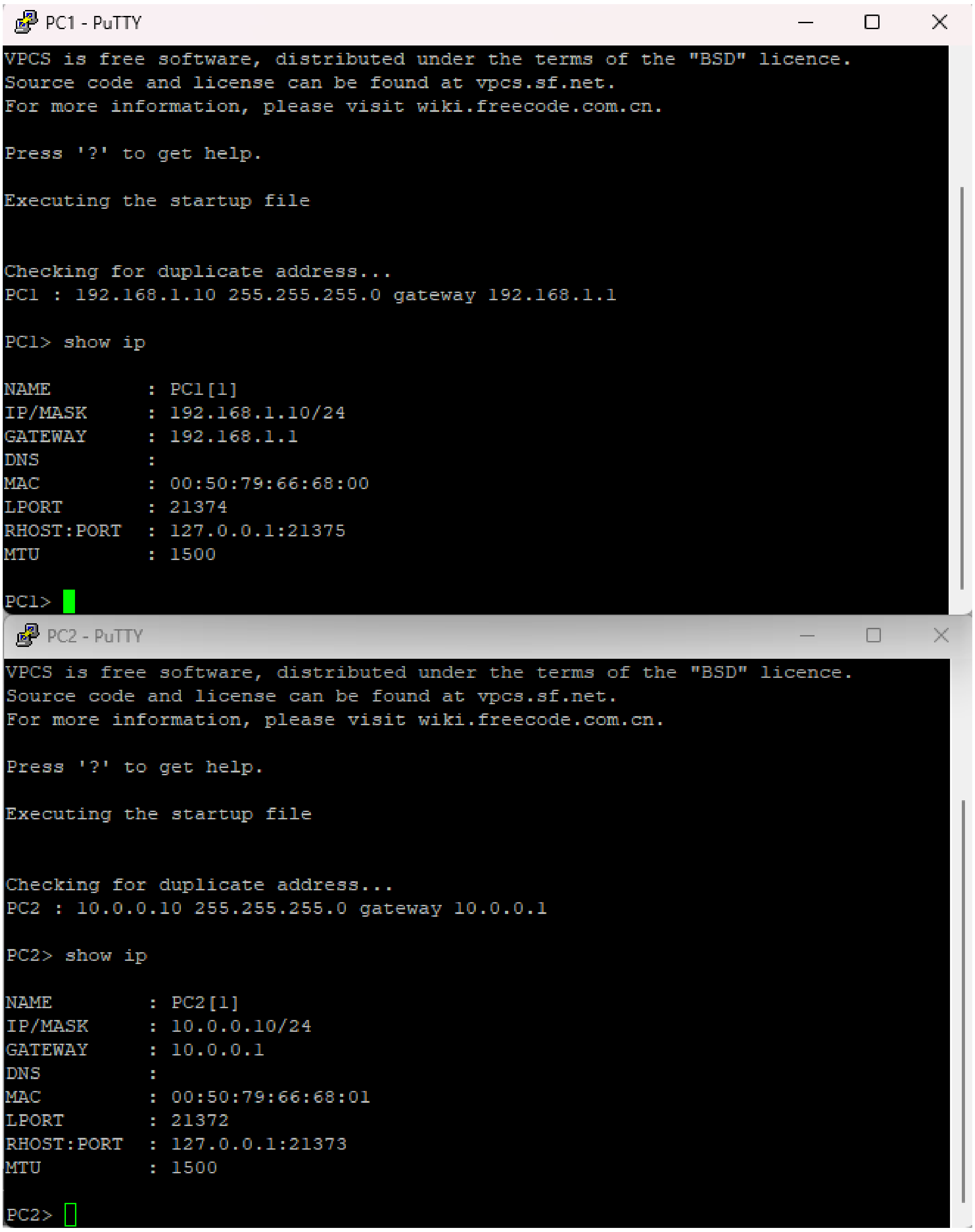
ip address 10.0.0.1 255.255.255.0

no shutdown

exit

end

write memory

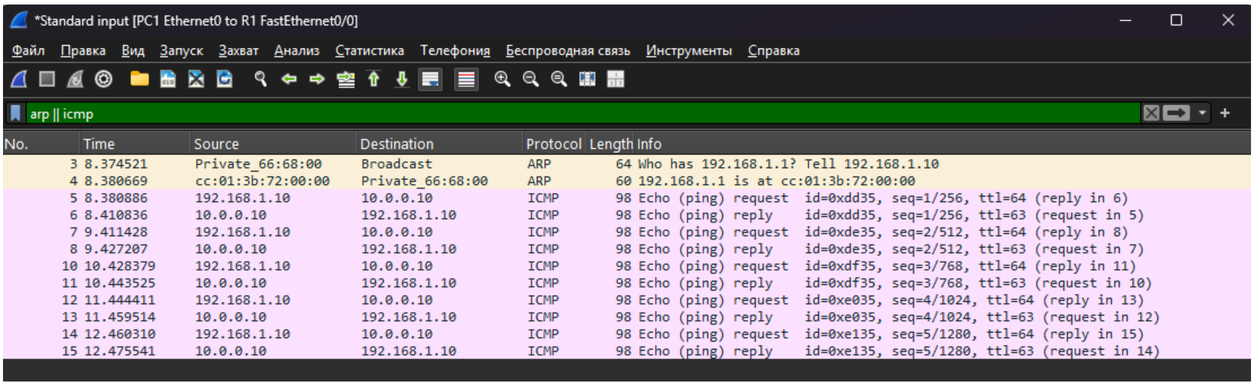


6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера.

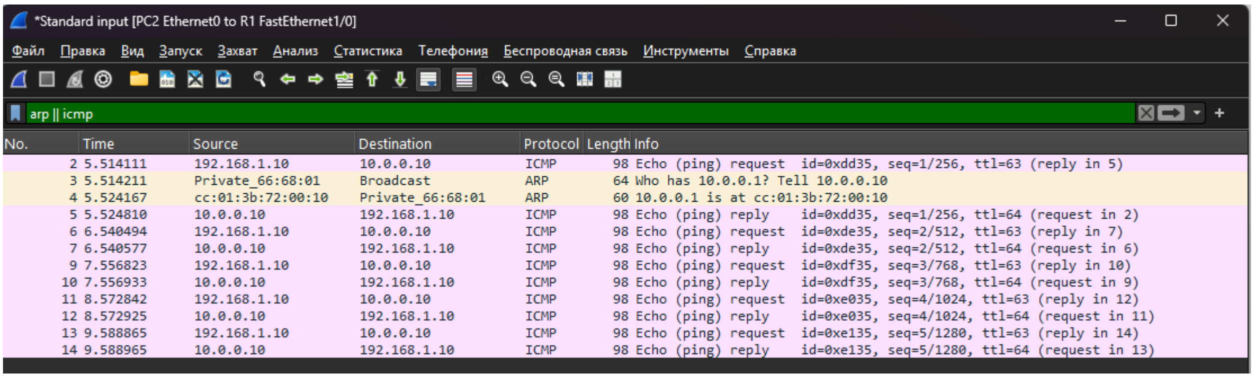
```
PC1> ping 10.0.0.10

10.0.0.10 icmp_seq=1 timeout
84 bytes from 10.0.0.10 icmp_seq=2 ttl=63 time=20.584 ms
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=15.134 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=15.025 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=15.499 ms

PC1>
```

7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark.

```
PC1> ping 10.0.0.10

84 bytes from 10.0.0.10 icmp_seq=1 ttl=63 time=30.028 ms
84 bytes from 10.0.0.10 icmp_seq=2 ttl=63 time=15.866 ms
84 bytes from 10.0.0.10 icmp_seq=3 ttl=63 time=15.229 ms
84 bytes from 10.0.0.10 icmp_seq=4 ttl=63 time=15.211 ms
84 bytes from 10.0.0.10 icmp_seq=5 ttl=63 time=15.321 ms

PC1>
```



No.	Time	Source	Destination	Protocol	Length	Info
3	8.374521	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.1? Tell 192.168.1.10
4	8.380669	cc:01:3b:72:00:00	Private_66:68:00	ARP	60	192.168.1.1 is at cc:01:3b:72:00:00
5	8.380886	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdd35, seq=1/256, ttl=64 (reply in 6)
6	8.410836	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdd35, seq=1/256, ttl=63 (request in 5)
7	9.411428	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xde35, seq=2/512, ttl=64 (reply in 8)
8	9.427207	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xde35, seq=2/512, ttl=63 (request in 7)
10	10.428379	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdf35, seq=3/768, ttl=64 (reply in 11)
11	10.443525	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdf35, seq=3/768, ttl=63 (request in 10)
12	11.444411	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe035, seq=4/1024, ttl=64 (reply in 13)
13	11.459514	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe035, seq=4/1024, ttl=63 (request in 12)
14	12.460310	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe135, seq=5/1280, ttl=64 (reply in 15)
15	12.475541	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe135, seq=5/1280, ttl=63 (request in 14)



No.	Time	Source	Destination	Protocol	Length	Info
2	5.514111	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdd35, seq=1/256, ttl=63 (reply in 5)
3	5.514211	Private_66:68:01	Broadcast	ARP	64	Who has 10.0.0.1? Tell 10.0.0.10
4	5.524167	cc:01:3b:72:00:10	Private_66:68:01	ARP	60	10.0.0.1 is at cc:01:3b:72:00:10
5	5.524810	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdd35, seq=1/256, ttl=64 (request in 2)
6	6.540494	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xde35, seq=2/512, ttl=63 (reply in 7)
7	6.540577	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xde35, seq=2/512, ttl=64 (request in 6)
9	7.556823	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xdf35, seq=3/768, ttl=63 (reply in 10)
10	7.556933	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xdf35, seq=3/768, ttl=64 (request in 9)
11	8.572842	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe035, seq=4/1024, ttl=63 (reply in 12)
12	8.572925	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe035, seq=4/1024, ttl=64 (request in 11)
13	9.588865	192.168.1.10	10.0.0.10	ICMP	98	Echo (ping) request id=0xe135, seq=5/1280, ttl=63 (reply in 14)
14	9.588965	10.0.0.10	192.168.1.10	ICMP	98	Echo (ping) reply id=0xe135, seq=5/1280, ttl=64 (request in 13)

ARP протокол:

Заголовки аналогичны пункту 4, но узнаём MAC-адрес шлюза.

ICMP Запрос от PC1 до R1:

```
Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -,
Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:3b:72:00:00 (cc:01:3b:72
    Destination: cc:01:3b:72:00:00 (cc:01:3b:72:00:00)
    Source: Private_66:68:00 (00:50:79:66:68:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 10.0.0.10
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x35dd (13789)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: ICMP (1)
    Header Checksum: 0x7910 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.10
    Destination Address: 10.0.0.10
    [Stream index: 0]
Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x42d5 [correct]
    [Checksum Status: Good]
    Identifier (BE): 56629 (0xdd35)
    Identifier (LE): 13789 (0x35dd)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 6]
    Data (56 bytes)
```

ICMP инкапсулирован в IP.

Source: Private_66:68:00 (00:50:79:66:68:00) -MAC-адрес PC1

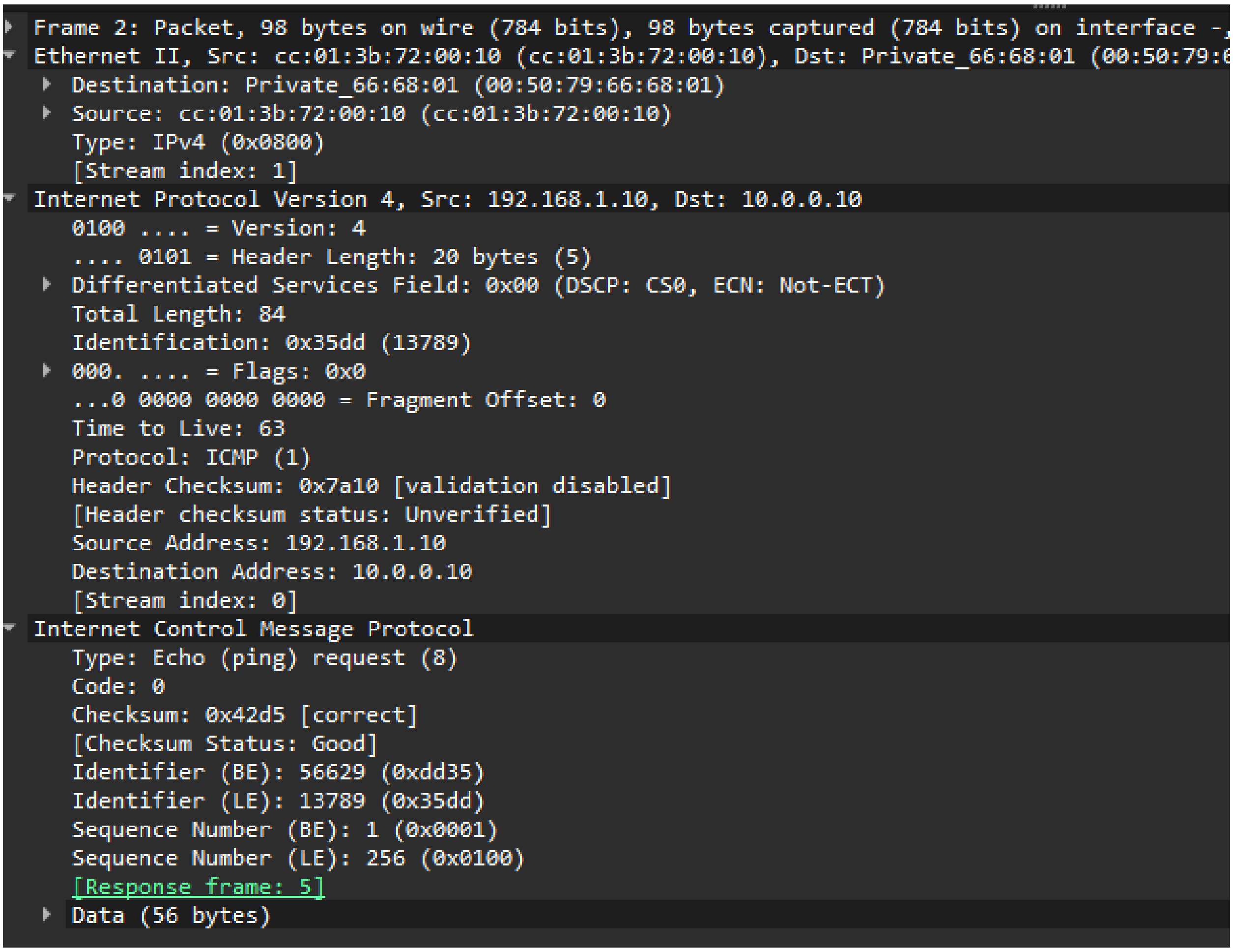
Destination: cc:01:3b:72:00:00 (cc:01:3b:72:00:00) - MAC адрес R1(маршрутизатор).

Source Address: 192.168.1.10 - ip PC1

Destination Address: 10.0.0.10 - ip PC2

Time to Live: 64

От R1 до PC2:



```
Frame 2: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -,
Ethernet II, Src: cc:01:3b:72:00:10 (cc:01:3b:72:00:10), Dst: Private_66:68:01 (00:50:79:6
    Destination: Private_66:68:01 (00:50:79:66:68:01)
    Source: cc:01:3b:72:00:10 (cc:01:3b:72:00:10)
    Type: IPv4 (0x0800)
    [Stream index: 1]
Internet Protocol Version 4, Src: 192.168.1.10, Dst: 10.0.0.10
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x35dd (13789)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 63
    Protocol: ICMP (1)
    Header Checksum: 0x7a10 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.10
    Destination Address: 10.0.0.10
    [Stream index: 0]
Internet Control Message Protocol
    Type: Echo (ping) request (8)
    Code: 0
    Checksum: 0x42d5 [correct]
    [Checksum Status: Good]
    Identifier (BE): 56629 (0xdd35)
    Identifier (LE): 13789 (0x35dd)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 5]
    Data (56 bytes)
```

Destination: Private_66:68:01 (00:50:79:66:68:01) -MAC-адрес PC2

Source: cc:01:3b:72:00:10 (cc:01:3b:72:00:10) - MAC адрес R1(маршрутизатор).

Source Address: 192.168.1.10 - ip PC1

Destination Address: 10.0.0.10 - ip PC2

Time to Live: 63 – снизился при прохождении маршрутизатора.